

DANA Article

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Abstract

1 Introduction

The floods that affected the Valencia region in October 2024 have been described as one of the most severe natural disasters in recent Spanish history, resulting in 228 fatalities and affecting thousands of people. In such contexts of uncertainty and disruption, news media play a crucial role in society, as they are responsible for disseminating information and shaping how events are publicly interpreted[1, 2]. For this reason media coverage contributes to the construction of meaning around disasters are understood and responded to.

1.1 Literature Review

1.2 A framework for Rhetoric Alarmism

2 Materials and Methods

2.1 Corpus Construction

This study analyzes a dataset of 1,340 headlines from different official news sources. The aim is to examine the presence of 'alarmist' framing in the coverage of the DANA event. The unit of analysis defined is the *journalistic headline* and in the case of social media, a sample for *n-words* is taken from the post. The study period covers from 29th October to 30 November 2024, separating the data into two periods: the first from 29th October to 4th November (*Impact*), and the second from 5th November to 30th November (*Post*). This division allows for a comparative analysis of the framing used in the immediate aftermath of the DANA event and in the subsequent weeks.

To ensure an appropriate representation of Spanish media ecosystem, the data set includes headlines from a variety of sources, which include national and local scopes. The data set includes headlines from the following media outlets:

Data collection followed a reproducible procedure based on structured search queries. Whenever possible, the *Advanced Google Search Engine* was used, combining boolean operators and site domains. When this method was insufficient and provided poor results, internal search engines of media sites were used specifically in social media. A limitation was encountered in the case of Instagram, as the platform currently does not provide a reliable date-filtering mechanism comparable to search engines or other social media platforms such as X (Twitter). This constraint makes it difficult to reproduce

Media Type	Media Outlet
National Newspapers	<i>20 minutos, ABC, AS, EFE, El Mundo, El País</i>
Local Newspapers	<i>EuropaPress, La Razón, La Vanguardia, Marca</i> <i>Las Provincias, Levante EVM, El Mediterráneo,</i> <i>Torrent al día, Valencia Plaza</i>
National Radio	<i>La Cope, La Ser, Onda Cero</i>
Local TV	<i>7TeleValencia, À Punt, LaOchoTv</i>
National TV	<i>Antena 3, Telecinco, La Sexta, RTVE, Cuatro</i>
Social Media	<i>X, Facebook, Tiktok</i>

the same sampling procedure, as result we decided to exclude this social media from the final corpus construction.

The general search pattern was as follows:

(dana OR "gota fría" OR lluvias OR torrenciales) AND Valencia AND 2024
site:<domain>

Explicit criteria were defined to ensure consistency across the headlines collected.

Inclusion Criteria

- Explicit reference to the phenomenon under study: DANA or heavy rainfall in Valencia
- Publication within the defined temporal window
- Headlines must originate from verifiable sources (URL or platform identifier)
- For social media: a minimum relevance threshold (≥ 1000 interactions, such as likes or shares)

Exclusion Criteria

- Irrelevant or ambiguous content not directly related to the event
- Duplicated entries or republished content
- News outside the temporal scope
- Non-comparable formats, lacking a clean headline structure (except for social media)

The sampling strategy consisted of selecting the first 50 relevant headlines per outlet, ordered according to their appearance in the search results and constrained by the predefined inclusion criteria. In cases where media outlets did not yield sufficient results within the defined temporal window, only those headlines meeting the inclusion criteria were retained. Therefore, the number of observations varies across outlets, with some sources contributing fewer than the intended 50 headlines.

Following the data collection process, a structured database was constructed in which each observation corresponds to an individual headline. The final corpus comprises 1347 headlines. For each observation, a set of variables was defined: the *media outlet* (source name), publication date, and the full headline text. Additionally, other categorical variables were incorporated, such as *media type* (e.g., press, radio, television, or social media) and the *temporal phase* of the event (impact or post-event stage). The dataset also includes an estimate of audience reach and the corresponding URL linking to the original content. The information related to audiences was collected from ...

2.2 Data Preprocessing

Text Normalization and Linguistic Processing Prior to analysis, a text standardization process was applied to ensure consistency and reduce noise. The cleaning procedure included converting all text to lowercase, replacing punctuation with whitespace, collapsing multiple whitespace characters into a single space, and removing residual special characters. For social media content, additional preprocessing steps were implemented, including the removal of hashtags (#), mentions (@), emojis, and other non-linguistic symbols.

Once the text was normalized, all headlines were lemmatized to improve pattern detection. This process transforms each word (token) into its canonical base form, thereby minimizing the impact of inflectional variations such as verb conjugations, plural forms, and gender agreement. Lemmatization was performed using the UDPipe library with a Spanish pre-trained model, which provides tokenization, part-of-speech tagging, and lemma extraction. Each headline was thus converted into a sequence of lemmas representing the original text. This lemmatized representation serves as the input for the subsequent regex-based detection process used to identify alarmist linguistic patterns.

3 Conclusions

We conclude that....

External data

The experimental dataset is available at....

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References

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